

Simulator Training in High-Performance Environments

A summary of findings from the scientific literature

This is a review of what the research says about simulator-based training in high-performance fields — aviation, military, surgical and medical teams, and elite sport. It does not draw conclusions about any specific application; it presents the findings briefly for us to reflect and discuss what parts we think should be implemented in how we choose and use the sim to get better results on the water.

Two questions organise the summary:

- 1. What aspects of using a simulator improve performance in real situations — for individuals and for teams?*
- 2. What is the best protocol — session duration, structure, and the brief / sim / debrief cycle — for translating sim training into real-world performance?*

Part 1 — What improves real-world performance

Individual performance

Deliberate practice, not exposure, drives improvement.

Skill gains come from structured, goal-directed practice with specific objectives and immediate feedback – not from time spent in the environment. Repetition without design produces diminishing returns and can cement errors. *(Ericsson, Krampe & Tesch-Römer, 1993.)*

Practice difficulty has an optimum.

Learning is fastest when tasks are hard enough to produce errors but not so hard that performance breaks down. Studies on error rates point to an optimum around a 15-30% error rate (roughly a 70-85% success rate) for efficient motor and cognitive learning. *(Wilson et al., 2019, on the "85% rule"; broader motor-learning literature.)*

Feedback type matters.

Positive feedback – indicating successful or better-than-average performance – tends to enhance motor learning more than negative feedback indicating poorer performance. *(Wulf & Lewthwaite, 2016, OPTIMAL theory.)*

Transfer depends on similarity and reflection.

What is learned in a simulator transfers to the real task most strongly when the simulated situation resembles the real one in the features that matter, and when the learner explicitly connects the simulated experience to the real-world behaviour. Physical and perceptual fidelity matter less than relevance to the decisions and processes being trained. *(Simulation transfer literature; Salas et al., 2009.)*

Team performance

Shared mental models are the strongest predictor of team performance.

Across team-performance research, the degree to which team members hold the same internal representation of a situation – a "shared mental model" (SMM) – is one of the most consistent predictors of coordinated performance, particularly under time pressure. Teams with strong SMMs outperform equally skilled teams with weak ones. *(Mathieu et al., 2000; DeChurch & Mesmer-Magnus, 2010, meta-analysis.)*

Shared mental models predict performance beyond communication.

In studies separating the two, shared understanding (SMM) predicts performance over and above closed-loop communication (CLC). Communication is the mechanism for distributing and confirming understanding; the

shared understanding itself is the more decisive factor. Without a shared model underneath, communication has less to contribute. *(Espevik et al., 2021.)*

Coordination shifts from explicit to implicit as shared models mature.

As teams develop strong shared models, they rely less on explicit verbal communication and more on anticipation – predicting each other's actions and needs without being told. This shift is associated with improved performance, especially under time pressure. *(Entin & Serfaty, 1999; Espevik, Johnsen & Eid, 2011.)*

Cross-training builds shared models.

Having team members learn each other's roles and responsibilities improves the shared model of how the team interacts, which in turn improves coordination and performance. *(Volpe et al., 1996; Marks et al., 2002.)*

Shared mental models take substantial time to build.

Shared understanding is a complex cognitive structure that develops through repeated shared experience. Short interventions are typically insufficient – one controlled study found an 8-hour team-training programme produced no measurable improvement in teamwork behaviours or shared understanding, attributed in part to the time required to build and share such models. Measurable development is more typically observed over weeks to a full season. *(Espevik et al., 2021; sport case studies, e.g. Gershgoren et al., 2013; Richards et al., 2025.)*

Part 2 — The best protocol

Session duration

Shorter sessions, capped by cognitive fatigue.

High-cognitive-load simulator work is limited by fatigue, which can degrade performance well within a long session. Evidence points to roughly 45-90 minutes of active simulation as the effective window, with around 2 hours (including debrief) as a practical maximum. *(Cognitive fatigue literature; distributed-training studies.)*

Frequency and spacing

Distributed practice beats massed practice.

One of the most robust findings in learning science: practice spread across multiple spaced sessions produces better long-term retention than the same amount of practice concentrated into fewer, longer sessions. Massed practice can show good immediate performance but poorer retention. *(Cepeda et al., 2006, meta-analysis of 317 experiments; Mawson & Kang, 2025.)*

Gaps between sessions support consolidation.

The interval between sessions – ideally including sleep – is when learning consolidates. Sessions packed too close together can interfere with each other; an interval of roughly 24 hours or more supports retention. *(Memory consolidation literature; McGaugh, 2000.)*

Structure: brief / sim / debrief

The debrief is the highest-value component.

Across the simulation literature, structured debriefing is consistently identified as the single most important element of a simulation-based learning experience – more important than the simulated activity itself. Reflection does not happen automatically; it requires dedicated time and facilitation. *(Simulation-based education literature.)*

Debriefs produce large, measurable performance gains.

A meta-analysis of 46 studies found properly conducted debriefs improved performance by an average of about 25%, with team-focused debriefs improving group performance by up to ~38%. The average debrief in these studies lasted around 18 minutes. Effective debriefs share four features: active self-learning, developmental (non-punitive) intent, reflection on specific events, and structure. *(Tannenbaum & Cerasoli, 2013.)*

Debriefing against an expert model builds shared understanding.

Debriefs structured around an expert or pre-defined model of good performance ("guided team self-correction") produce more accurate and more shared mental models than unstructured or purely chronological debriefs. *(Smith-Jentsch et al., 2008.)*

Debrief timing.

Debriefs are most effective when conducted close in time to the simulated experience, while the events are fresh and engagement is high. *(Simulation-based education literature.)*

One objective per session.

Sessions focused on a single clear objective outperform sessions attempting to improve many things at once; spreading focus dilutes the learning. *(Elite-sport case studies, e.g. Ashford et al., 2023.)*

A documented "sensemaking" dip.

When teams begin actively rebuilding a shared model, coordination can temporarily decline before improving, as the existing shared understanding is renegotiated. This dip is a documented part of the development process rather than a sign of failure. *(Richards et al., 2025.)*

Summary of key findings

- Improvement comes from **deliberate, structured practice with feedback**, not exposure or volume.
- For individuals, **practice difficulty has an optimum** (~70–85% success), and **positive feedback** aids learning.
- **Transfer** to the real task depends on relevance of the simulated situation and explicit connection to real-world behaviour — not on physical fidelity.
- For teams, **shared mental models are the strongest predictor of performance**, and predict performance **beyond communication** alone.
- Shared models take **substantial time** to build (weeks to a season); short interventions tend to fail.
- The best protocol is **shorter (45–90 min), spaced sessions** rather than long, occasional ones.
- The **debrief is the most valuable component**, producing large measurable gains, and is strongest when structured, developmental, close in time, and built around a model of good performance.
- **One objective per session**; expect a temporary **dip** when shared understanding is being rebuilt.

Principal sources

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